

Exercise 2: Mathematical Background

1 Vector and Tensors

(1) Use the Levi-Civita tensor to establish the following identities:

- a) $\vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{A} \cdot \vec{B})\vec{C}$
- b) $\vec{A} \cdot (\vec{B} \times \vec{C}) = (\vec{A} \times \vec{B}) \cdot \vec{C}$
- c) $\vec{\nabla}(uv) = u\vec{\nabla}v + v\vec{\nabla}u$
- d) $\vec{\nabla} \cdot (u\vec{A}) = u\vec{\nabla} \cdot \vec{A} + \vec{A} \cdot \vec{\nabla}u$
- e) $\vec{\nabla} \times (u\vec{A}) = u\vec{\nabla} \times \vec{A} + \vec{\nabla}u \times \vec{A}$
- f) $\vec{\nabla} \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot \vec{\nabla} \times \vec{A} - \vec{A} \cdot \vec{\nabla} \times \vec{B}$
- g) $\vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A}$
 where $\nabla^2 \vec{A} = \frac{\partial^2 \vec{A}}{\partial x^2} + \frac{\partial^2 \vec{A}}{\partial y^2} + \frac{\partial^2 \vec{A}}{\partial z^2}$
- h) $\vec{\nabla}(\vec{A} \cdot \vec{B}) = \vec{A} \times (\vec{\nabla} \times \vec{B}) + (\vec{A} \cdot \vec{\nabla})\vec{B} + \vec{B} \times (\vec{\nabla} \times \vec{A}) + (\vec{B} \cdot \vec{\nabla})\vec{A}$
- i) $\vec{\nabla} \times (\vec{A} \times \vec{B}) = (\vec{B} \cdot \vec{\nabla})\vec{A} - (\vec{A} \cdot \vec{\nabla})\vec{B} + \vec{A}(\vec{\nabla} \cdot \vec{B}) - \vec{B}(\vec{\nabla} \cdot \vec{A})$
- j) $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0$
- k) $\vec{\nabla} \times (\vec{\nabla}f) = 0$

(2) Let $\vec{r}$ be a vector from the origin to the point (x, y, z) and let $\vec{r}'$ be a vector from the origin to the point (x', y', z') . Evaluate the following expressions in terms of $\vec{r}$ and $\vec{r}'$.

- a) $\frac{d\hat{r}(t)}{dt}$
 where $\hat{r} = \frac{\vec{r}}{|\vec{r}|}$ is the unit vector, in the direction of $\vec{r}$.
- b) $\vec{\nabla} |\vec{r} - \vec{r}'|$
- c) $\vec{\nabla} \frac{1}{|\vec{r} - \vec{r}'|}$
- d) $\vec{\nabla} \frac{1}{|\vec{r} - \vec{r}'|^2}$
- e) $\vec{\nabla}^2 \frac{1}{|\vec{r} - \vec{r}'|}$, for $\vec{r} \neq \vec{r}'$
- f) $\vec{\nabla} \times \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3}$, for $\vec{r} \neq \vec{r}'$

(3) Prove the following identity for a vector field $\vec{v}$

$$\int_V (\vec{\nabla} \times \vec{v}) dV = \int_S (\hat{n} \times \vec{v}) dA$$

4 The trace of a second-rank tensor is the sum of all its diagonal elements. That is, $\text{Tr } \mathbf{T} = \sum_{i=1}^3 T_{ii}$. Show that $\text{Tr} \mathbf{T}$ is invariant under rotation.

5 Show that the sum of squares of all elements of a second-rank tensor is invariant under rotation.

6 If $\mathbf{S}$ and $\mathbf{T}$ are two second-rank tensors, show that $\sum_{ij} S_{ij} T_{ij}$ is invariant under rotation.

2 The Dirac δ -function and Fourier Transforms

(7) The sequence of Gaussian functions $D_n(x) = \sqrt{\frac{n}{\pi}} e^{-nx^2}$, $n = 1, 2, \dots$, in the limit $n \rightarrow \infty$, represents a $\delta(x)$ -function. Show that

- a) $\int_{-\infty}^{\infty} \delta(x) g(x) dx = g(0)$
- b) $\delta(x) g(x) = \delta(x) g(0)$

(8) Consider the step function, $\theta(x) = 1$ for $x > 0$ and $\theta(x) = 0$ for $x < 0$. Show that $\frac{d\theta}{dx} = \delta(x)$.

(9) Show that

$$\delta(y(x)) = \sum_i \frac{\delta(x-x_i)}{\left| \frac{dy}{dx} \right|_{x=x_i}}.$$

(10) Compute the Fourier transform of the δ -function and the Gaussian function $D_1(x)$.

(11) If $f(x)$ and $g(k)$ are Fourier transforms of each other, show that

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-\infty}^{\infty} |g(k)|^2 dk$$

sometimes called the Parseval's theorem.

(12) Using the Dirac δ -functions in appropriate coordinates, express the following charge distributions as three-dimensional charge densities $\rho(\vec{r})$.

- a) In spherical coordinates, a charge Q uniformly distributed over a spherical shell of radius R .
- b) In cylindrical coordinates, a charge λ per unit length uniformly distributed over a cylindrical surface of radius b .
- c) In cylindrical coordinates, a charge Q spread uniformly over a flat circular disc of negligible thickness and radius R .
- d) The same as in (c), but using spherical coordinates.

3 Gradient, Divergence, Curl

(13) Recall that we defined the gradient of a scalar function as a vector derivative which points in the direction in which the scalar undergoes the maximum change. The cylindrical coordinates (ρ, θ, z) , defined by

$$x = \rho \cos \theta, \quad y = \rho \sin \theta, \quad z = z$$

are functions of the Cartesian coordinates. By computing the gradients of the cylindrical coordinates, find the unit vectors $\hat{\rho}$, $\hat{\theta}$, $\hat{z}$.

(14) Repeat the exercise to find the unit vectors in spherical coordinates defined by

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta$$

in terms of the Cartesian coordinates.

(15) Calculate $|\vec{r} - \vec{r}'|$ in Cartesian, cylindrical, and spherical coordinates.

(16) Let ξ_1, ξ_2, ξ_3 be three generalised, curvilinear coordinates in three-dimensional Euclidean space. An infinitesimal line element in these coordinates is given by the general expression

$$dl^2 = g_{11}(\xi_1, \xi_2, \xi_3) d\xi_1^2 + g_{22}(\xi_1, \xi_2, \xi_3) d\xi_2^2 + g_{33}(\xi_1, \xi_2, \xi_3) d\xi_3^2$$

where the g_{ij} are components of the metric tensor. Find the components of the metric tensor in Cartesian, cylindrical, and spherical coordinates.

(17) Use the above result to write down, in each of the coordinate systems,

- a) the differential distances along each coordinate direction
- b) the differential areas obtained by holding one coordinate fixed
- c) the infinitesimal volume element

(18) The expressions for the gradient, divergence, curl, and Laplacian in the general, curvilinear coordinates are given by

$$\begin{aligned}\vec{\nabla}U &= \hat{\xi}_1(g_{11})^{-1/2}\frac{\partial U}{\partial \xi_1} + \hat{\xi}_2(g_{22})^{-1/2}\frac{\partial U}{\partial \xi_2} + \hat{\xi}_3(g_{33})^{-1/2}\frac{\partial U}{\partial \xi_3} \\ \vec{\nabla} \cdot \vec{F} &= (g_{11}g_{22}g_{33})^{-1/2} \times \\ & \left[\frac{\partial}{\partial \xi_1} \{F_1(g_{22}g_{33})^{1/2}\} + \frac{\partial}{\partial \xi_2} \{F_2(g_{33}g_{11})^{1/2}\} + \frac{\partial}{\partial \xi_3} \{F_3(g_{11}g_{22})^{1/2}\} \right] \\ \vec{\nabla} \times \vec{F} &= \hat{\xi}_1(g_{22}g_{33})^{-1/2} \left[\frac{\partial}{\partial \xi_2} \{F_3(g_{33})^{1/2}\} - \frac{\partial}{\partial \xi_3} \{F_2(g_{22})^{1/2}\} \right] \\ & + \hat{\xi}_2(g_{33}g_{11})^{-1/2} \left[\frac{\partial}{\partial \xi_3} \{F_1(g_{11})^{1/2}\} - \frac{\partial}{\partial \xi_1} \{F_3(g_{33})^{1/2}\} \right] \\ & + \hat{\xi}_3(g_{11}g_{22})^{-1/2} \left[\frac{\partial}{\partial \xi_1} \{F_2(g_{22})^{1/2}\} - \frac{\partial}{\partial \xi_2} \{F_1(g_{11})^{1/2}\} \right] \\ \nabla^2 U &= (g_{11}g_{22}g_{33})^{-1} \times \\ & \left[\frac{\partial}{\partial \xi_1} \left\{ (g_{22}g_{33}/g_{11})^{1/2} \frac{\partial U}{\partial \xi_1} \right\} + \frac{\partial}{\partial \xi_2} \left\{ (g_{33}g_{11}/g_{22})^{1/2} \frac{\partial U}{\partial \xi_2} \right\} + \frac{\partial}{\partial \xi_3} \left\{ (g_{11}g_{22}/g_{33})^{1/2} \frac{\partial U}{\partial \xi_3} \right\} \right]\end{aligned}$$

Use the expressions for the unit vectors and components of the metric tensor derived earlier to compute the above expressions in various coordinates.